

Exploring Bayesian inference methods for flood-risk management

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710036384

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Abstract

In the world of risk management, decision makers have to be aware of potential costs with each action taken. Bayesian inference can allow stakeholders to learn from large volumes of data to create confidence intervals for predicted outcomes and help weigh trade-offs between false positives and false negatives. In this report, I investigate multiple Bayesian approximation methods - Stochastic Variational Inference (SVI), Markov Chain Monte Carlo (MCMC) and Laplace Approximation - and evaluate their performance on a flood-risk dataset. Each method's implementation is described in detail to show how intractable posteriors can be approximated; highlighting trade-offs in computational complexity, uncertainty calibration, and predictive performance. The results discuss how each approach scales to real-world data and how the choice of method depends on operational constraints, risk tolerance and the need for timely predictions.

I certify that all material in this document which is not my own work has been identified.

1 Introduction

Bayesian inference is a powerful statistical tool that allows us to draw conclusions from data, whilst accounting for uncertainty. It relates the fundamental components of Bayes' Theorem to continually refine predictions. In the realm of machine learning, neural networks are model architectures designed to optimise complex objective functions, traditionally relying on deterministic weight functions for point estimates of model parameters. However the extreme flexibility of these networks leaves them susceptible to overfitting, whereas Bayesian inference finds the best explanation for our model's parameters. Bayesian Neural Networks (BNNs) treat weights as distributions rather than fixed values, enabling a natural quantification of uncertainty and also incorporating prior knowledge into the learning process. This is particularly advantageous for many real-world applications, more specifically flood risk management and analysis. Flooding is caused by a myriad of factors that are inherently dynamic and uncertain in nature, therefore by adopting a Bayesian framework, BNNs provide not only point estimates but also full predictive distributions. Given the severe consequences of underestimating flooding severity, BNNs benefit decision-makers with estimates of the confidence or uncertainty associated with predictions, making it an invaluable tool for risk assessment, improved decision making and resource allocation in emergency planning. The first step for any Bayesian computation is to derive the joint distribution of all the unknown variables given our observed data. Then for performing computations, approximation inference methods are employed due to the intractability of obtaining exact posterior distributions. The two methods this report will focus on are Variational Approximation (VA) and Markov Chain Monte Carlo (MCMC). Variational approximation is a method for estimating the full posterior distribution with guaranteed convergence under certain conditions. This is achieved by minimising the Kullback-Leibler (KL) divergence between the approximate distribution and the true posterior. Its computational efficiency and ability to scale to data very well makes it suitable for our high-frequency flood data. The latter sampling-based method, MCMC, remains an important tool for estimating statistical models, particularly when high accuracy in uncertainty quantification is required. It works by constructing a Markov chain whose stationary distribution is the true posterior allowing for asymptotically exact samples. Both VA and MCMC seek to approximate the same intractable posterior distribution in BNNs - VA does so by optimising over a restricted family of distributions, whereas MCMC uses stochastic sampling to capture the full complexity of the posterior. This trade-off comes at the cost of significantly higher computational demands and slower convergence, depending on the dimensionality level of parameter spaces. By leveraging approximation methods with the flexibility and rigor of BNNs, this report will explore how these advanced techniques can be applied to flood risk management and ultimately assess the effectiveness of these methods in capturing uncertainty, thereby provide more insight into flood risk assessment and mitigation strategies.

2 Methods

2.1 Variational Approximation

The first method, Variational approximation (VA), expresses the estimation of the posterior distribution $p(\theta | D)$ as an optimisation problem; approximating the posterior density through a simpler distribution. We measure the closeness of this inference turned optimisation problem using the Kullback-Leibler (KL) divergence:

$$\text{KL}(q(\theta) \| p(\theta | D)) = \int q(\theta) \log \frac{q(\theta)}{p(\theta | D)} d\theta.$$

This divergence measures the information loss when $q(\theta)$ is used to approximate $p(\theta | D)$; and so a lower KL divergence means a closer match between the two distributions.

Minimising the KL divergence is the same as maximising the Evidence Lower Bound (ELBO), which is derived as follows. Starting from the marginal likelihood:

$$\log p(D) = \log \int p(D, \theta) d\theta,$$

we introduce the variational distribution $q(\theta)$ and apply Jensen’s inequality to the concave function (logarithm), which allows us to obtain a tractable lower bound on the intractable integral:

$$\log p(D) = \log \int q(\theta) \frac{p(D, \theta)}{q(\theta)} d\theta \geq \int q(\theta) \log \frac{p(D, \theta)}{q(\theta)} d\theta \equiv \mathcal{L}(q).$$

And so, the ELBO is given by:

$$\mathcal{L}(q) = \mathbb{E}_{q(\theta)}[\log p(D | \theta)] - \text{KL}(q(\theta) \| p(\theta)).$$

In practice, we optimize the parameters of $q(\theta)$ by maximizing $\mathcal{L}(q)$ using stochastic gradient methods. One of the most common choices of stochastic gradient methods is Adam (Adaptive moment estimation), due to its adaptive learning rate and robustness to noisy gradients. The key enabler for this optimisation is the reparameterisation trick. When sampling from a parameter-dependent distribution $q(\theta; \varphi)$ that is typically non-differentiable with respect to φ ; the reparameterisation trick resolves this by expressing the sampled parameter θ as a deterministic function of φ and an independent noise variable ϵ drawn from a fixed distribution (usually Gaussian). This transforms the originally non-differentiable sampling operation into one that is smoothly dependent on the variational parameters, essentially making it feasible to apply a standard neural network training algorithm, like Adam, to complex Bayesian models. Under the mean-field assumption, we can simplify the variational approximation by factorising the joint distribution $q(\theta; \varphi)$ into a product of independent distributions for each parameter dimension:

$$q(\theta; \phi) = \prod_{i=1}^K q_i(\theta_i; \phi_i)$$

where K represents the total number of parameters in the network. Whilst the mean-field assumption and reparameterisation trick greatly reduce the computational complexity to facilitate scalable variational inference, it comes at the cost of ignoring correlation between parameters; this simplification can often lead to an underestimation of the true posterior uncertainty.

Similar to the mean-field approach, one powerful application of variational approximation in deep learning is the Variational Autoencoder (VAE).[2] VAEs combine neural networks with a latent variable model. Instead of compressing data through deterministic encoders, it opts for a generative framework, where the encoder, otherwise known as the ‘inference’ model, outputs the mean $\mu_\phi(x)$ and log-variance $\log \sigma_\phi^2(x)$ for each input x . The generative part of the model, decoder, maps the latent codes back into the observed data space. This formulation of a differentiable function from the encoder, allows VAEs to be trained end-to-end with an optimisation method like Adam. Its objective of maximising the ELBO function, utilises the KL divergence as a regulariser to ensure that the latent space is structured and that the learned representations do not deviate too far from the prior distribution.

Variational Approximation offers a unified framework for efficiently handling intractable posteriors, whether in simple Bayesian models or in more complex neural architectures like Variational Autoencoders (VAEs), through the optimisation of the Evidence Lower Bound function (ELBO). This method marries computational feasibility with the probabilistic perspective on model parameters making it a favourable choice of method for regression tasks where precise estimates of risk are paramount.

2.2 Markov Chain Monte Carlo

Markov Chain Monte Carlo (MCMC) methods represent the canonical approach for full Bayesian inference, opting for a sampling-based approach to approximating intractable posteriors. Rather than relying on deterministic transformations seen in Variational approximation, MCMC simulates draws from the posterior by constructing a Markov chain whose stationary distribution is the true posterior $p(\theta \mid D)$. Once the chain converges, a sufficient number of samples can be used to approximate posterior expectations or intervals.

The most prominent MCMC algorithms are Metropolis-Hastings and Gibbs sampling. In Metropolis-Hastings, parameter updates θ^* are drawn from a proposal distribution $q(\theta^* \mid \theta^{(t)})$ and are either accepted or rejected based on the ratio:

$$\alpha = \frac{p(\theta^* \mid D) q(\theta^{(t)} \mid \theta^*)}{p(\theta^{(t)} \mid D) q(\theta^* \mid \theta^{(t)})}.$$

If $\alpha \geq 1$, θ^* is accepted; otherwise, θ^* is accepted with probability α . The advantage of Metropolis-Hastings is that it recovers the true posterior distribution given infinite iterations, thorough mixing and its ability to evaluate $\log p(\theta \mid D)$.

Gibbs sampling takes a different route - sequentially sampling from the full conditional distributions of each parameter given all of the others.[1] This approach is often convenient when each of these conditional distributions is easier to sample from than the joint distribution itself. The Gibbs sampler iterates through each component of the parameter vector

$$\begin{aligned} \theta_1^{(t+1)} &\sim p(\theta_1 \mid \theta_2^{(t)}, \dots, \theta_d^{(t)}, D), \\ &\dots \\ \theta_d^{(t+1)} &\sim p(\theta_d \mid \theta_1^{(t+1)}, \dots, \theta_{d-1}^{(t+1)}, D). \end{aligned}$$

By cycling through the parameters and conditioning on the most recently updated values of the others, the algorithm converges to the true posterior.

Both methods generate posterior samples and in theory can both yield asymptotically exact inferences given enough iterations and good chain mixing. Gibbs works best when each parameter's conditional distribution is tractable, which is ideal for hierarchical models with semi-conjugate priors. On the other hand, M-H can be used when conditionals aren't in recognisable families, although it can require careful tuning of step size or the proposal covariance. If well-tuned it can efficiently traverse multi-modal posteriors. The choice is dependent on model structure, conjugacy and dimensionality.

An essential tool in MCMC is to ensure that samples generated from the Markov chain reliably represent the target distribution with convergence diagnostics. Traceplots visualise sampled values from the parameter space across iterations to observe whether the chain has reached a stationary region. A 'caterpillar'-like appearance is indicative of good convergence and distinct phases suggest that the chain is stuck in local modes. Another tool is the autocorrelation measure which shows the correlation between samples at different lags. Ideally, samples from a well-mixed chain should be nearly independent, a low autocorrelation. A high autocorrelation would indicate that more iterations are needed for a reliable estimate of the posterior. The Gelman-Rubin diagnostic, also known as the potential scale reduction factor, runs multiple chains with different starting values; comparing the variance between chains to the variance within each chain. Values greater than 1 indicate that the chains have not converged, signaling further iterations are needed.

2.3 Laplace Approximation

An alternative method to the computationally heavy MCMC is Laplace approximation, which is often more practical when dealing with large models or datasets. It centers around linearising the log-posterior distribution near its maximum a posteriori (MAP) estimate, thereby providing a local Gaussian approximation to the true posterior. Instead of sampling from the full parameter space, the Laplace method starts with a standard training procedure to obtain a set of parameters and then approximates the Hessian of the log-posterior around the point. Once the Hessian (H) is estimated, the approximate covariance $\Sigma \approx (H + \lambda I)^{-1}$ is then used to define a Gaussian in parameter space. So by sampling from $\mathcal{N}(\hat{\theta}_{\text{MAP}}, \Sigma)$ we gain insight into how sensitive the network’s predictions are to small perturbations in its weights, a measure of uncertainty. Since Laplace approximation requires the full Hessian of the negative log-posterior at ..., it can often become numerically unstable so several approximation strategies exist. Diagonal approximation is where only the diagonal entries of the Hessian are retained, meaning each parameter’s variance is estimated independently. This approach is straightforward as it ignores correlations between different weights and biases. A more expressive middle ground is to store some low-rank factors of the Hessian via Low-Rank Approximation, which captures some inter-parameter correlations. A ”last-layer Laplace” is a practical variant that approximates only the Hessian for the final network layer to significantly reduce the dimension of the curvature matrix, thus consolidating a pragmatic middle ground for large datasets.

Each iteration of MCMC, whether M-H or Gibbs, requires the evaluation of the log-likelihood of all data points, which can become computationally intensive for large datasets. Storage and manipulation of large design matrices can further strain workstation environments, which can force us to resort to subsampling strategies - selecting random mini-batches of data to approximate gradients or acceptance ratios. Sitting somewhat in between VA & MCMC, Laplace approximation sidesteps the heavy iterative sampling MCMC requires[3], yet sacrificing flexibility in capturing multi-modal posteriors. In contrast, VA’s transformation of the posterior inference problem into an optimisation task by stochastic gradient methods allows for efficient scaling to large datasets. While MCMC methods can capture complex posterior structures with careful tuning of proposal covariances in M-H or block updates in Gibbs, VA typically offers a faster convergence. Ultimately, the choice of inference method for flood-risk analysis hinges on trade-offs: MCMC offers deeper insights into the posterior’s shape at a high computational cost, VA prioritises speed and scalability through optimisation and Laplace approximation provides a practical compromise.

3 Experimental Design and Results

3.1 Stochastic Variational Inference

The flood dataset provided for this study contains multiple features relevant to flood risk; including continuous variables (e.g., elevation, slope), categorical variables (e.g. soil and building type) and three hazard-related columns – HAZARD 1 IN 30, HAZARD 1 IN 100 and HAZARD 1 IN 1000. These three columns were combined by taking their maximum value to create a single numeric target variable, ’flood severity’. As part of the data cleaning and preprocessing part, the following steps were taken: Firstly, categorical columns were converted into integer codes so that the neural network can handle them as numeric inputs. Then missing values in numeric columns were replaced with their corresponding column means. The dataset was then randomly split into the following subsets: training (70%), validation(15%) and testing (15%) to enable unbiased performance estimation. Considering the large raw coordinate values in our dataset, a StandardScaler was applied to all input columns so that each feature had zero mean and unit variance, mitigating any potential numerical instability.

For the neural network’s architecture, I designed a feed-forward network with two hidden layers. The first hidden layer has 50 neurons, each performing a linear transformation of the input features plus a bias term, outputting:

$$\mathbf{h}_1 = \text{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1)$$

where $\mathbf{W}_1 \in \mathbb{R}^{50 \times d}$ is the weight matrix, $\mathbf{b}_1 \in \mathbb{R}^{50}$ is a 50-dimensional bias vector and $\mathbf{x}$, the input vector. The ReLU (Rectified Linear Unit) activation helps the network capture complex relationships whilst mitigating the vanishing gradient problem often seen with traditional activations such as sigmoid or tanh. The second layer, which also uses 50 neurons, receives the output of the first hidden layer $\mathbf{h}_1$ for the linear transformation:

$$\mathbf{h}_2 = \text{ReLU}(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2)$$

The architecture’s output layer then uses a single unit to output a continuous value $\hat{y}$ representing the predicted flood severity.

For transforming this deterministic architecture into a Bayesian neural network, I began by placing priors on our parameters; each layer’s weights and biases received Normal priors with zero mean and unit variance, matching the shape of these priors with the corresponding layer’s parameter shape. Then drawing from a Uniform distribution, I introduced a single noise term, σ , to represent the homoscedastic noise in the observed target values.

Using Pyro’s AutoDiagonalNormal guide for a mean-field approximation, Stochastic Variational Inference (SVI) was employed to replace the intractable true posterior with a simpler diagonal Gaussian distribution $q(\theta; \phi)$ over parameters. The training process initialises the guide $q(\theta; \phi)$, and the reparameterisation trick allows gradients to propagate through the sampling process. The objective of the SVI loop is to maximize the Evidence Lower Bound (ELBO), which is defined as:

$$\mathcal{L}(\phi) = \mathbb{E}_{q(\theta; \phi)} [\log p(D \mid \theta)] - \text{KL}(q(\theta; \phi) \parallel p(\theta))$$

3.2 SVI results

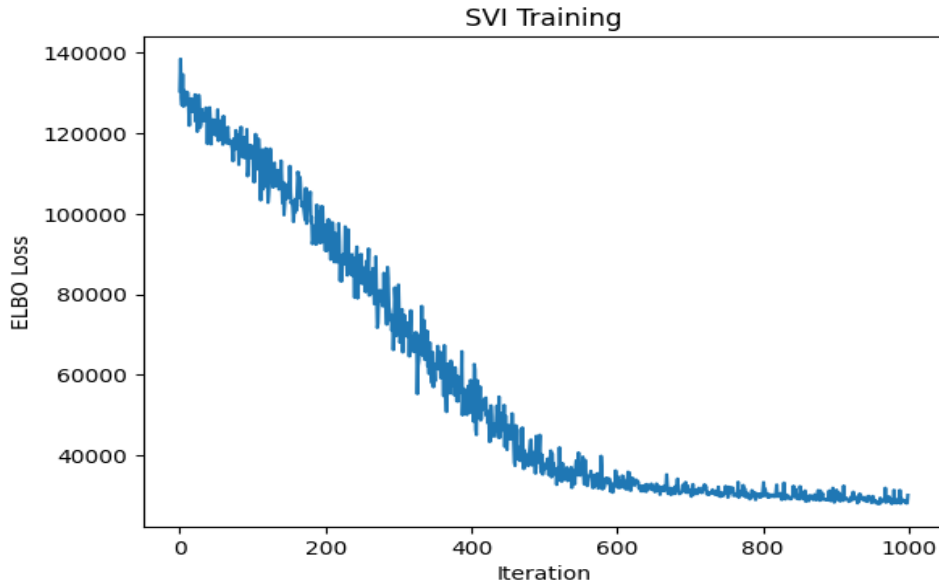


Figure 1: ELBO curve during SVI training

The SVI loop ran for 1000 iterations; monitoring the ELBO loss where a steadily decreasing ELBO indicates that the variational posterior becomes a better approximation of the true posterior.

The decline of the curve in Figure 1 is not perfectly smooth, highlighting the stochastic nature of mini-batch updates, but it’s overall shape demonstrates that the model progressively captured more information from the data whilst reducing the KL divergence.

After training, the model was evaluated on the test set, yielding a moderate performance with metrics:

- **RMSE (Root mean squared error):** 0.38
- **MAE (Mean absolute error):** 0.14
- **R²:** 0.45

3.3 Hamiltonian Monte Carlo

The next method explored is a variant of Hamiltonian Monte Carlo (HMC), called No-U-Turn Sampling (NUTS). It’s a powerful MCMC technique that leverages gradient information to make more informed proposals. Unlike traditional MCMC approaches, NUTS dynamically adapts the number of leapfrog steps, allowing for efficient exploration of high-dimensional parameter spaces. Given the high class imbalance in flood severity labels, the original regression task was reformulated into a binary classification problem to capture flood risk patterns. For this classification framework, I defined a new binary target variable so that a flood risk is indicated by a value of 1 when the flood severity is greater than zero, and 0 otherwise. To account for the large class imbalance, I undersampled the majority class (no flood risk) and incorporated class weights directly in the Bernoulli likelihood to assign higher importance to flood risk cases. The BNN’s architecture was designed as follows: a feedforward neural network with five hidden layers, each utilising 50 hidden neurons to introducing non-linearity. The final layer consisted of a single neuron which output the logits, representing the probability of flood risk, which is then passed through a Bernoulli likelihood to model the probability as a binary classification outcome.

For inference, NUTS was selected as the MCMC kernel. Priors of $\mathcal{N}(0, 0.1)$ were placed on each weight and bias, and step size for leapfrog integration was set to 0.003, with a target acceptance probability of 0.9 to ensure effective exploration of the posterior distribution. Moreso, 600 warmup steps were used to allow the samples were drawn to estimate the parameter distributions reliably. To optimise performance, specifically for the F1 score, a metric used to measure how well a model performs, a threshold search was conducted iteratively on the test set predictions, ensuring that the final model balanced precision and recall effectively.

3.4 No-U-Turn Sampling results

The performance of this classification model was evaluated with key metrics, including accuracy, F1-score and the receiver operating characteristic (ROC) and precision-recall (PR) curves. After various balancing strategies and hyperparameter tuning, the best classification metrics yielded were:

Model	Accuracy	F1 Score	ROC AUC	PR AUC
BNN (NUTS, Classification)	0.63	0.70	0.74	0.68

The table of results indicate that the five-hidden-layer BNN correctly identified a large portion of the minority ”flood risk” class with a high recall of 0.87, meaning that the model correctly identifies a substantial proportion of true flood-risk instances.

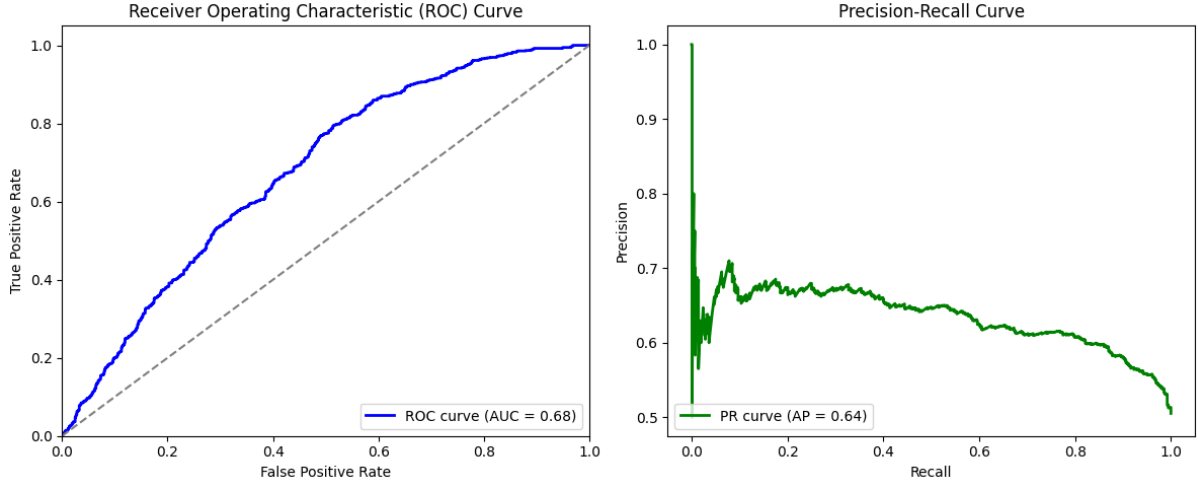


Figure 2: ROC and PR curve for NUTS

The corresponding precision of 0.59 reflects some trade-off in the form of false positives. However, such a trade-off can be acceptable in this case if the cost of a missed flood is far greater than the cost of a false alarm. With further refinements to the decision threshold, policymakers can utilise this to balance precision and recall according to their risk tolerance. The ROC and PR curve underscore the model’s discriminative ability across varying probability thresholds; our produced ROC shows a decent separation between positive and negative classes, whereas the PR curve highlights how precision drops at higher recall levels - mirroring the 0.59 precision at a recall of 0.87.

3.5 Laplace Approximation

The final method explored for flood-risk analysis is Laplace Approximation; to begin with, the data set was split into 70% training and 30% validation. Given the rarity of flood events in our data, a weighted cross-entropy was applied to ensure that errors on the minority class were penalised more heavily. I retained the same architecture for SVI and trained for 100 epochs. The final weights at the end of training are taken as the MAP estimate $\hat{\theta}_{\text{MAP}}$, which are stored as parameters for subsequent Laplace approximation. Two variants of Laplace approximation were investigated. In the first variant, I applied a diagonal Hessian approximation on the last layer only. This approach assumes that the majority of the uncertainty in the network’s predictions are concentrated in the final classification layer. As mentioned earlier, this reduces the computational burden whilst still capturing the local curvature at the decision boundary. For the second variant, I computed the diagonal Hessian approximation across all layers of the network to capture uncertainty not only in the final layer but also in the feature extraction stages of the network. 10 samples were drawn per validation pass, each yielding a set of predictive probabilities and then averaged to obtain an ensemble prediction.

3.6 Laplace Approximation results

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Final-Layer Laplace	0.935	0.488	0.202	0.285	0.881	0.326
All-Layers Laplace	0.771	0.174	0.684	0.278	0.881	0.358

Table 1: Comparison of Laplace Approximation results on flood-risk classification for two Hessian approaches: all-layers vs. final-layer.

Both approximation methods achieved relatively high accuracy, specifically final layer Laplace with 0.935. While this final-layer approach excelled in maintaining high precision (0.488) and overall accuracy, it suffered from a recall of just 0.202, indicating that nearly 80% of actual flood events went undetected. In contrast, the all-layers variant substantially increased recall to 0.684, albeit at the expense of both precision (0.174) and accuracy (0.771). In regards to the context of flood-risk management, a false negative (failing to detect a real flood) is far costlier than raising a false alarm, therefore the all-layers approach can be more favourable if minimising missed floods is a priority. However, if reducing false alarms is more critical then the final-layer Laplace may be preferred. For future research, a Kronecker-factorised method could be implemented for the Hessian to offer a balance between the diagonal and all-layer extremes to improve both recall and precision. At the cost of extra computational overhead it would allow for more reliable flood-risk detection in practice.

4 Conclusion

Our investigation of varying approximation methods used in conjunction with Bayesian neural networks entails how each different technique handles the intractable posterior distribution. The straightforward SVI used for Bayesian regression proved its capability in speed and scalability, however it struggled to approximate complex posterior structures. Further refinements could explore more expressive variational families to better capture parameter correlations and non-Gaussian behaviour. For a flood-risk classification problem, Laplace provided a flexible post hoc approach, offering local uncertainty approximations with moderate overhead. Evaluation of the two variants’ results expressed the trade-offs in flood-risk management that the cost of false negatives outweigh those of false positives. The NUTS method, representing a full MCMC sampling approach, demonstrated robust posterior exploration. The high F1 score of 0.70 suggested good handling of precision and recall. Its ability to capture complex dependencies among parameters and calibrate uncertainty well makes it stand out as a preferred method for uncertainty estimation, despite its computational intensity. Ultimately, the invaluable tool of Bayesian inference for a risk classification task as such is to employ a method that can provide decision-makers trustworthy uncertainty estimates. The choice however is fundamentally a question of balancing the latter with computational feasibility.

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